

LIRUI WANG

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EDUCATION

University of Washington, Seattle, WA (GPA:3.97 / 4.00)

Sep 2016 - Present

Bachelor of Science, Electrical Engineering, (Honors), College of Engineering

Bachelor of Science, Computer Science, (Honors), College of Arts and Sciences

Bachelor of Science, Mathematics, (Minor), College of Arts and Sciences

RESEARCH PROJECTS

Motion and Grasp Planning

March 2019 - Present

- Implement a network that combines data-driven and traditional method to synthesize grasp pose
- Prototype RL, IL, and MPC with PyBullet environment to map observed points directly to 6D pose increments in task space
- Jointly optimize motion and grasp planning in cluttered scene by extending the CHOMP and online learning frameworks

DeepIM Tracking

June 2018 - March 2019

- Tested DeepIM on YCB datasets and achieved robust results trained on synthetic data only
- Improved network training stability, pose distribution and occlusion generation
- Extended DeepIM in real-time tracking for occluded scene in YCB and built AR demos
- Worked on robot kinematics to build generalized vision system that can track robot arm

Flow and Depth Study

March 2018 - June 2018

- Investigated a few deep learning methods results for jointly 3D geometry predictions such as depths, flow, and egomotion
- Explored unsupervised structures and ideas such as SfMlearner and MonoDepth and analyzed them on a robotic setting

Baxter Manipulation

Aug 2017 - March 2018

- Worked on a manipulation task with PoseCNN, which enables Baxter to manipulate daily objects
- Experimented with camera models, robot kinematics, 3D geometry and ROS packages for tracking and control

WORK EXPERIENCE

Seattle, WA

Sep 2019 - Present

Computer Vision Teaching Assistant, UW CSE 455

Shenzhen, China

Jun 2017 - Aug 2017

Robotics Engineer Intern, DJI Technology

- Worked on autonomous robot battle which uses vehicles with TX1 onboard and DJI Manifold drones with Guidance
- Implemented functions such as take-off, tag detection, and refilling to localize drone and communicate
- Applied SLAM to mapping and localize, and built a top-level state machine to strategize and navigate the robot

LEADERSHIP AND INVOLVEMENT

Seattle, WA

Sep 2016 - Feb 2018

Computer Vision Lead

Advanced Robotics at the University of Washington

- Managed a team of 10 students to design computer vision tools for our robots in a competition RoboMasters hosted by DJI.
- Performed system integration of controls, navigation and serial communication

PERSONAL PROJECTS

- **Sound Scope:** Designed a sound processing pipeline from scratch for modelling the acoustics of a train whistle
- **Music Editor:** Created a demo app that can record sound pieces and translate into sheet music display UI using DTFT, and it was awarded Overall 1st place out of 35 teams in Seattle

HONORS

- Annual Deans List, University of Washington, *2016 - 2020*
- Tau Beta Pi Scholarship *2017 - 2019*
- Lawrence & Lucille Frey Endowed Electrical Engineering Scholarship *2017 - 2018*
- Patricia G. Lynch and Theodora & Eugene Russell Memorial Computer Science Scholarship *2018 - 2019*

COURSES

- **Computer Science:** Data Structure and Parallelism, Algorithms, Computer Vision, Introduction to Artificial Intelligence, Deep Learning, Modern Algorithm Toolbox, Graphics, Robotics, Robust Machine Learning Algorithms (Grad)
- **Electrical Engineering:** Digital Systems, Control System, Control Capstones, Filter Analysis, Discrete Time Linear System
- **Math:** Multivariable Calculus, Probability and Statistics, Numerical Analysis, Linear Analysis, Convex Optimization and Analysis (Grad), Topology, High Dimensional Statistics and Statistical Learning (Grad), Statistical Learning: Modeling, Prediction, And Computing (Grad)
- **English:** Technical Writing, Advanced Technical Writing in Electrical Engineering Statistics: Statistical Learning: Modeling, Prediction, And Computing (Grad)

SKILLS

- **Software Development Tools:** ROS, Ubuntu, ARM, Arduino, Nvidia TX Series, STM32
- **Programming Language:** Java, C++/C, Python, Matlab
- **Tools:** Git, CMake, OpenCV, DJI Guidance, Bash, Latex, Tensorflow, Pytorch, MXNet, OpenGL, CUDA, Qt
- **Rapid Prototyping:** 3D Printing, Laser Cutting, Machining, Soldering

EXTRACURRICULARS

- **Mens Pickup Soccer Team:** Played regularly at pickup soccer and joined the competition organized by UW Intramural Activities Center.
- **Math Club:** Joined the meetings to solve problems with other students in the group.